

Introduction to Neuroscience and Neural Engineering

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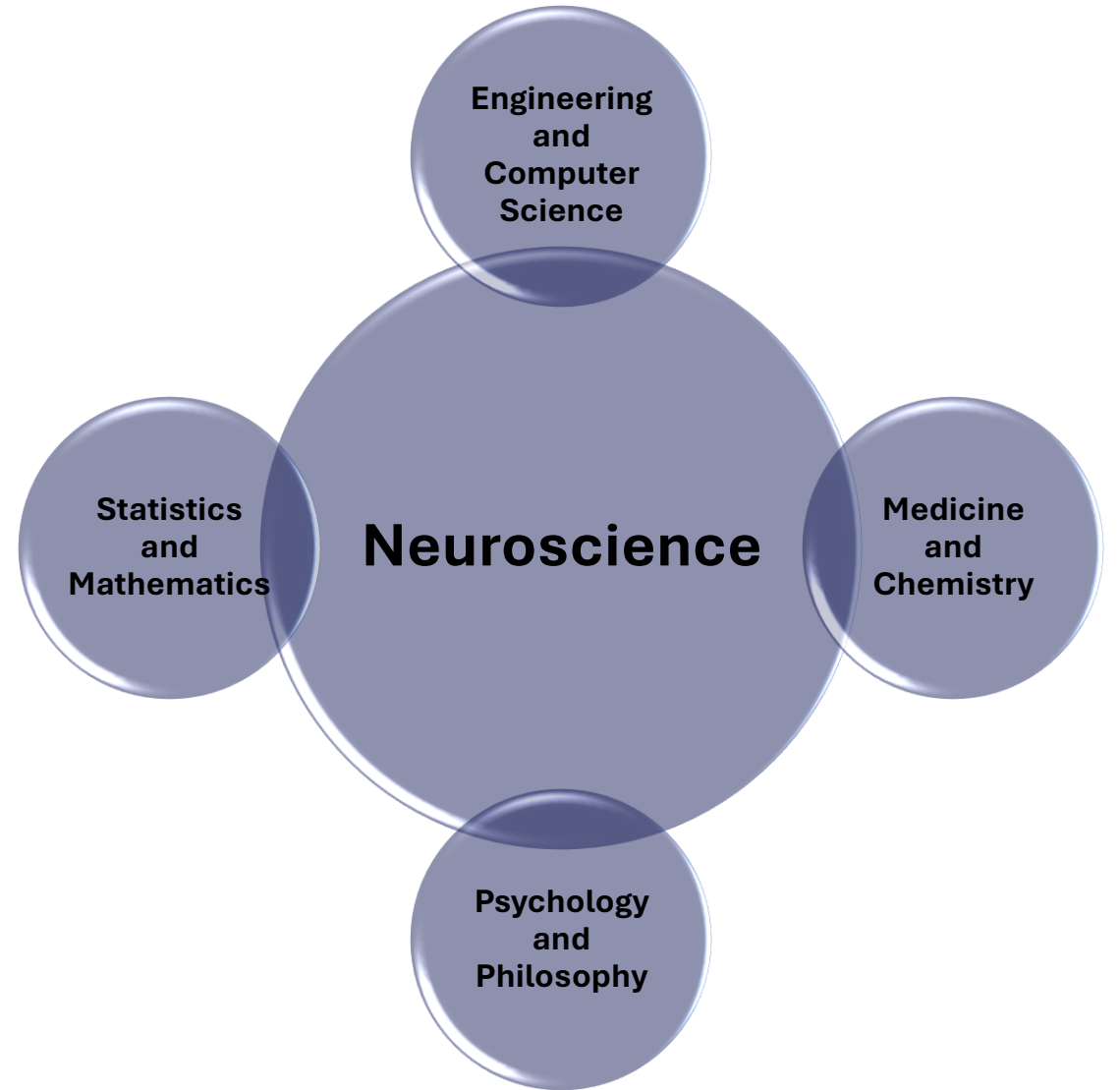
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What is neuroscience?

Neuroscience is the study of how the nervous system develops, its structure, and its impact on behavior and cognitive functions.

Neuroscience also examines neurological, psychiatric, and neurodevelopmental diseases.

Neuroscience's main scientific elements



Neuroscience's main branches

**Affective
Neuroscience**

**Cognitive
Neuroscience**

**Behavior
Neuroscience**

**Cultural
Neuroscience**

**Computational
Neuroscience**

**Cellular
Neuroscience**

**Systems
Neuroscience**

**Sensory
Neuroscience**

**Clinical
Neuroscience**



Neuroscience popular specializations

Neuroeconomics

Neuropathology

Neuropharmacology

Neurophysiology

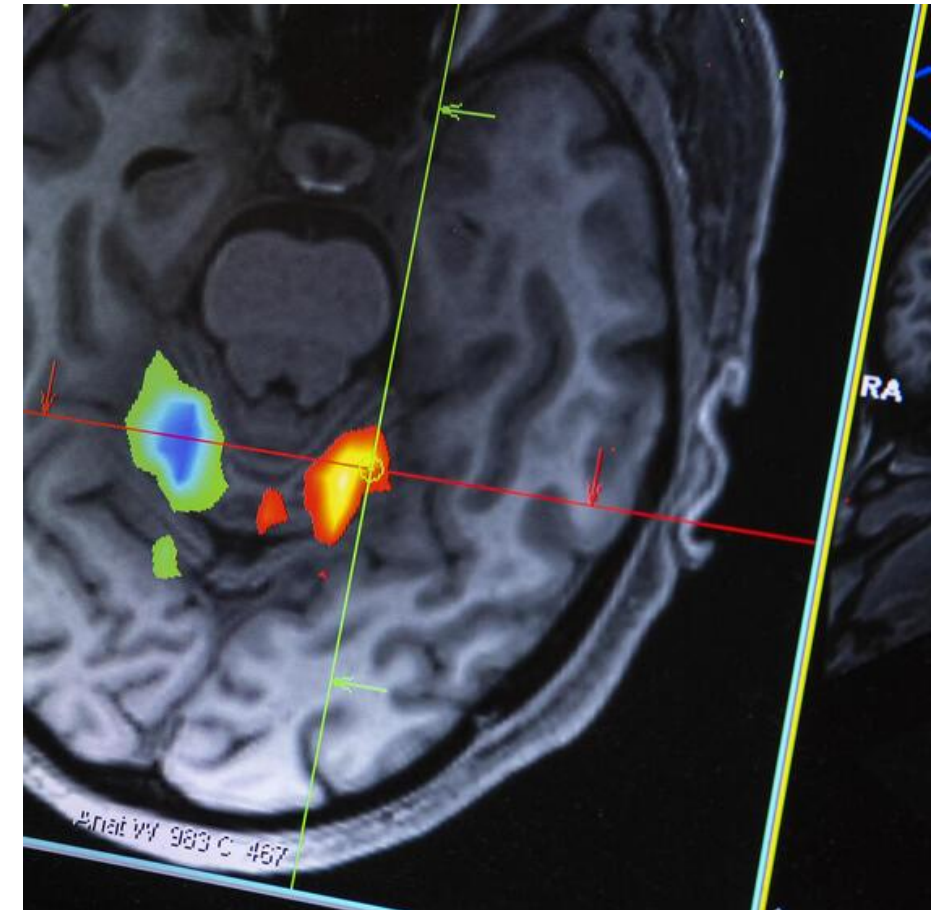
Neuropsychology

Neuroinformatics

Neuroimaging

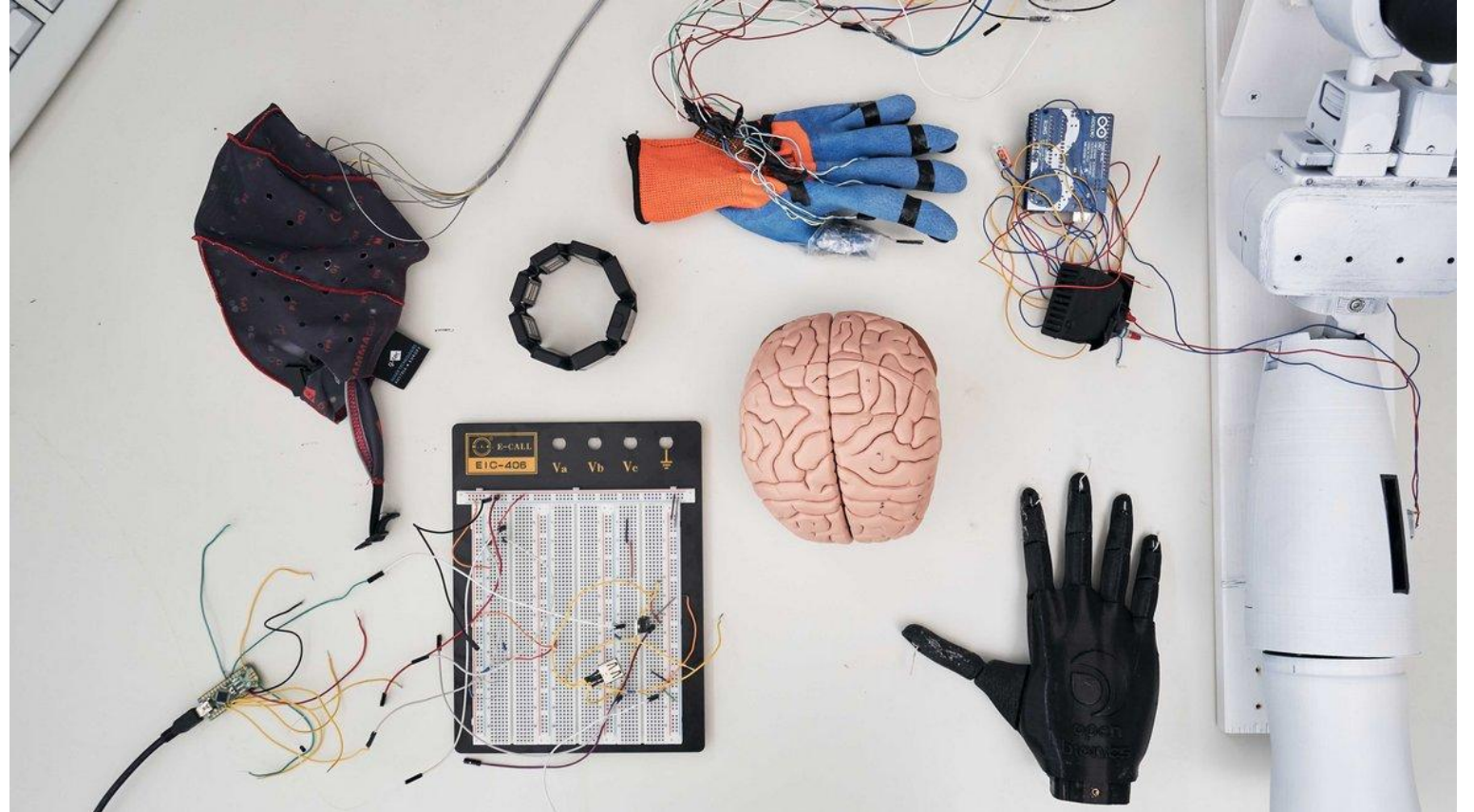
Neurohacking

NeuroAI

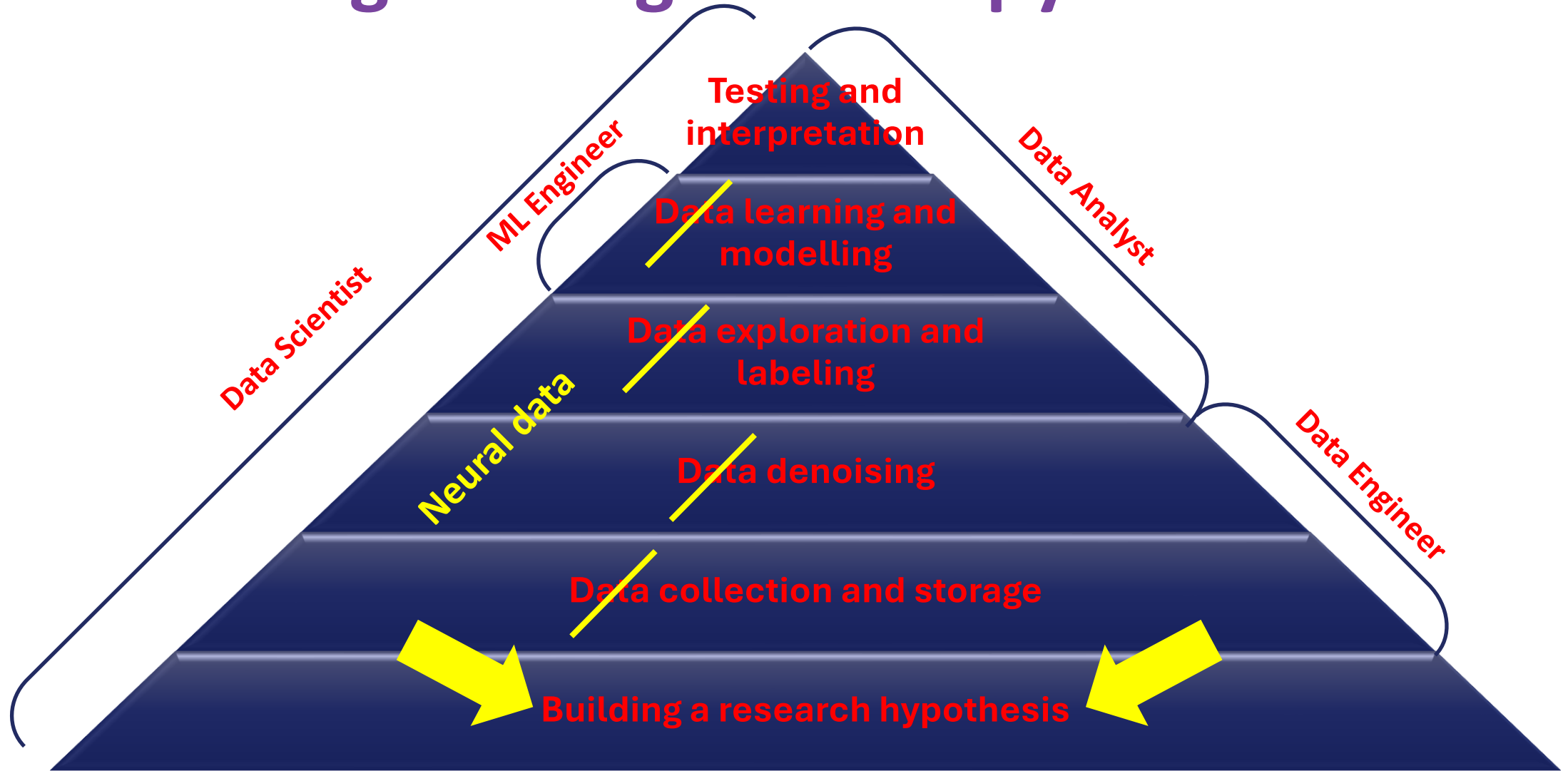


What is neuro-engineering?

Neuro-engineering uses engineering techniques to understand, repair, or upgrade brain systems.



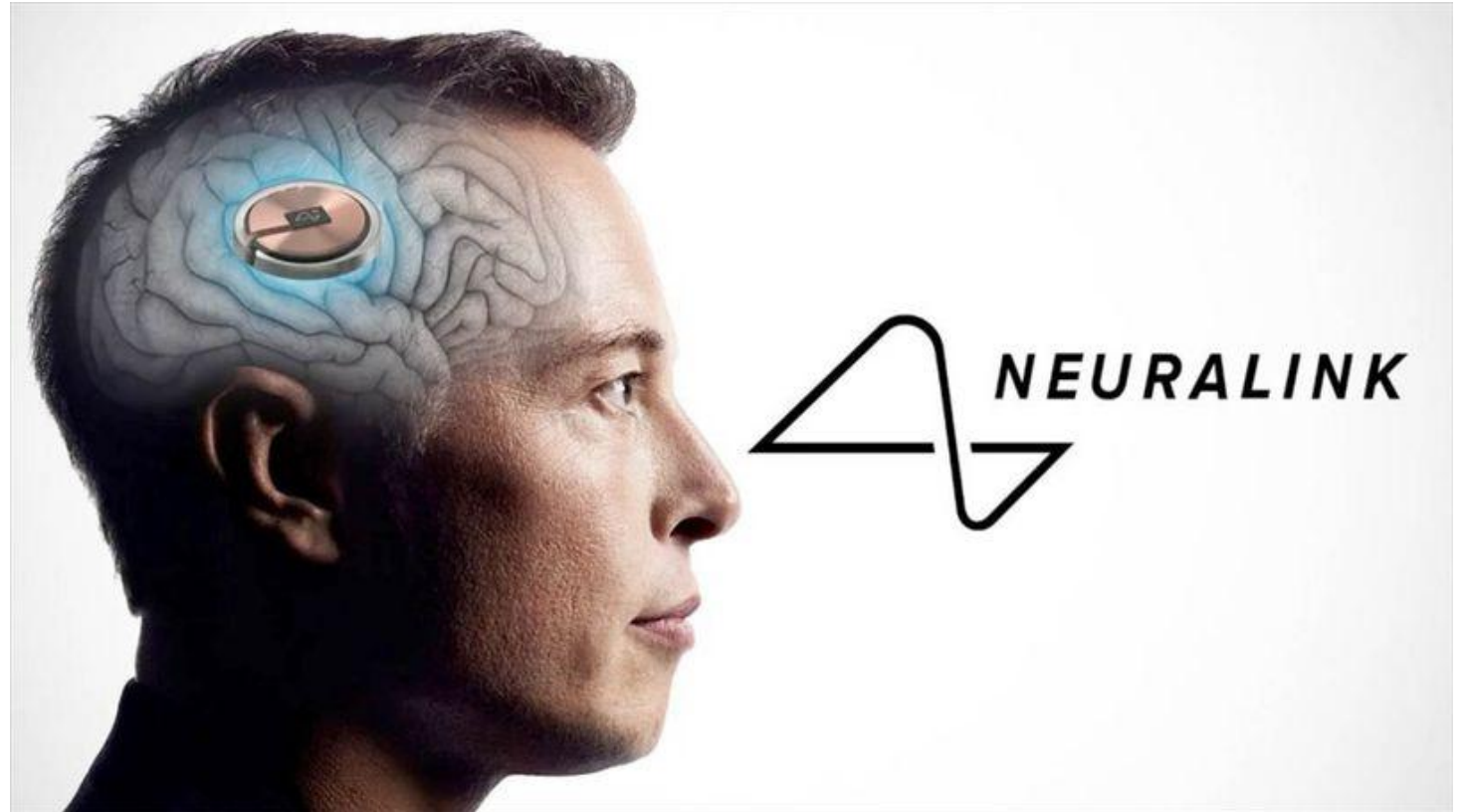
The data engineering wisdom pyramid



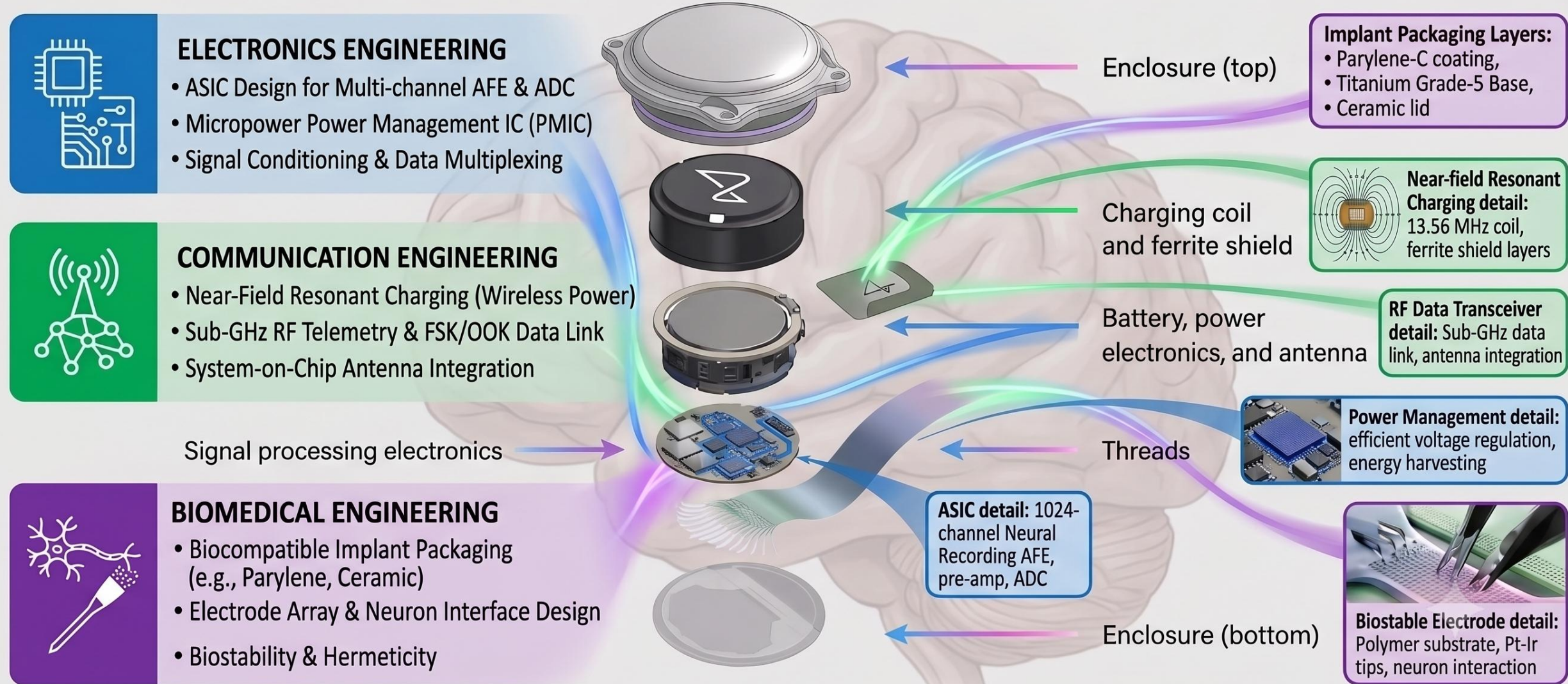


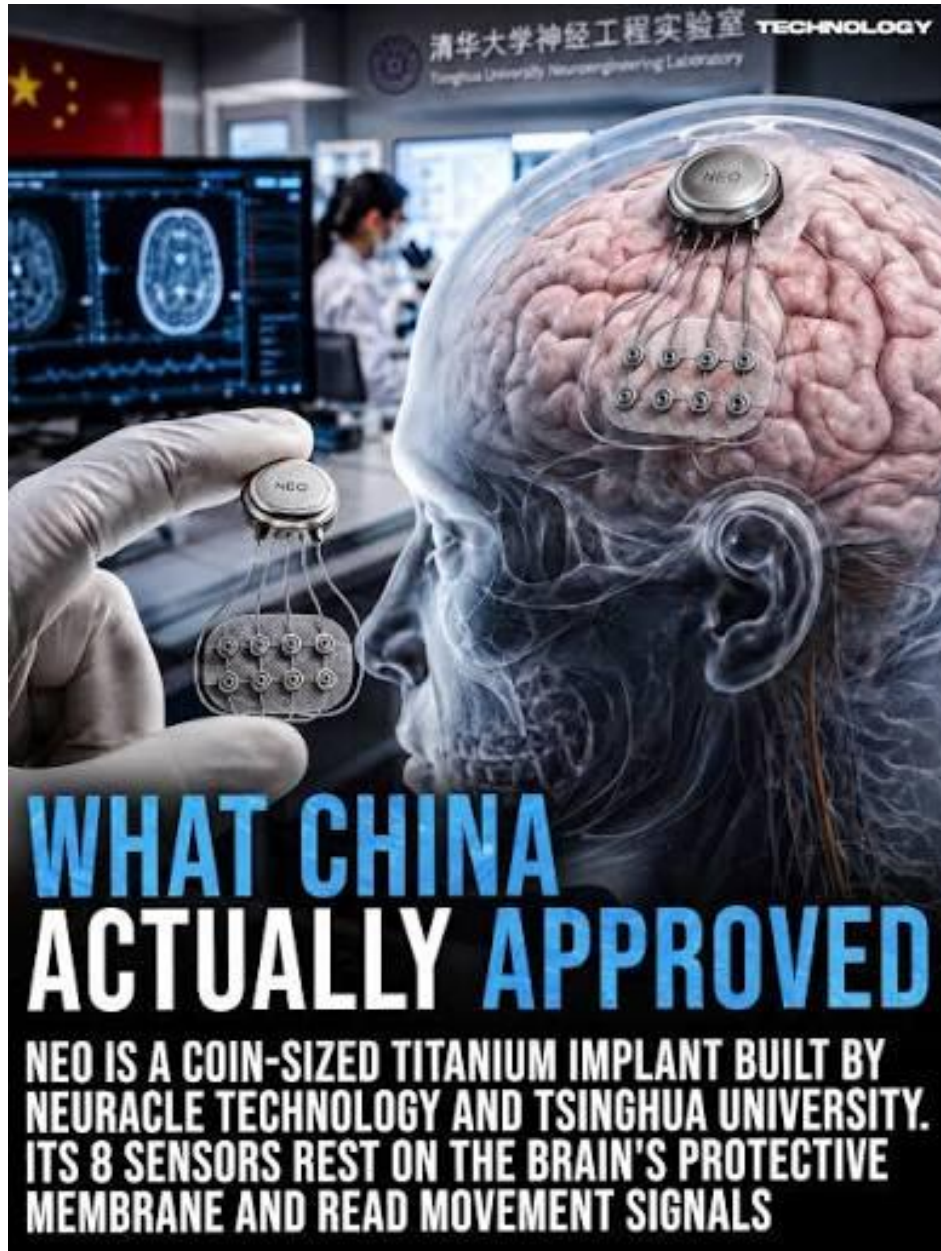
Careers at Neuralink

<https://neuralink.com/careers/>



Neuralink brain implant architecture: An Interdisciplinary Engineering Specialization

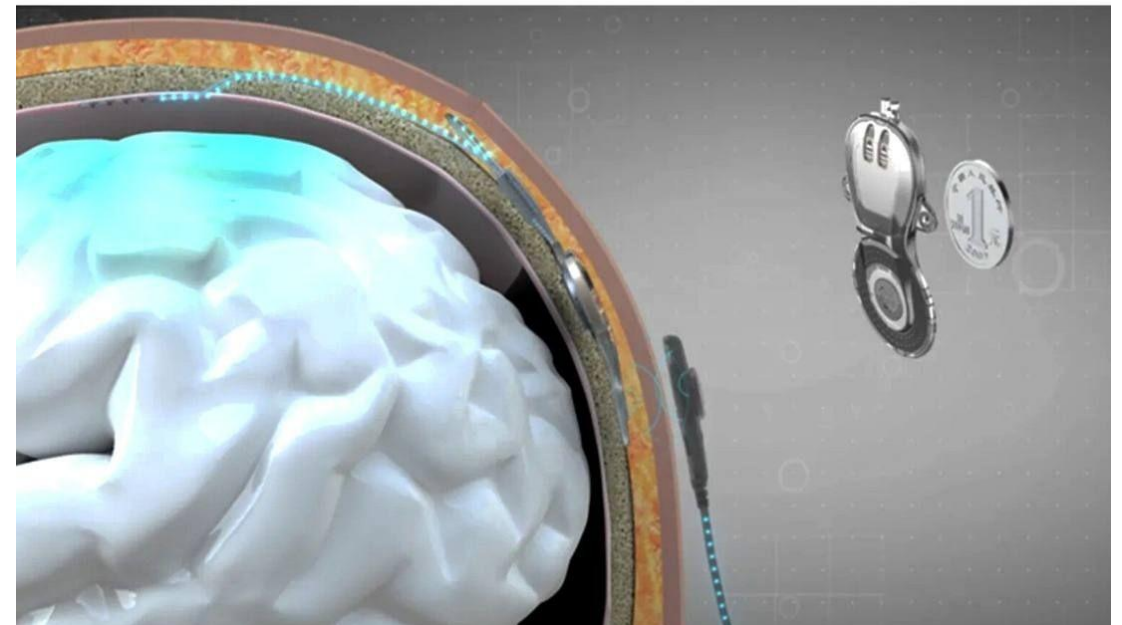




China just put its first commercial brain implant into a patient. It's smaller than Neuralink's, on purpose.

Surgeons in Shanghai have carried out the world's first commercial brain-computer interface surgery, using a coin-sized chip that sits on the brain rather than piercing it

July 17, 2026 - 7:49 am



What is a brain-computer interface (BCI)?

BCIs are communication systems that **bypass** the brain's normal output pathways of peripheral nerves and muscles.

Applications

Replace

- Prosthetic control
- Speller programs

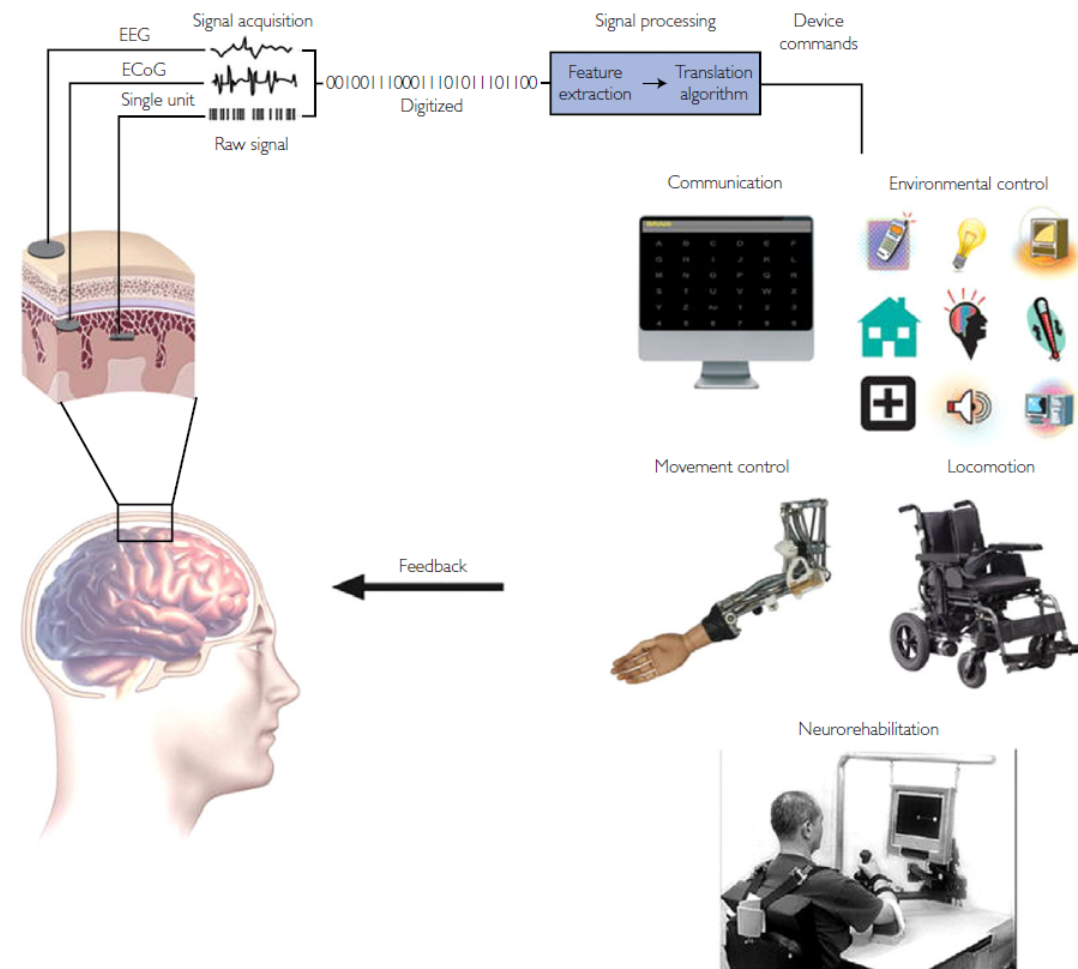
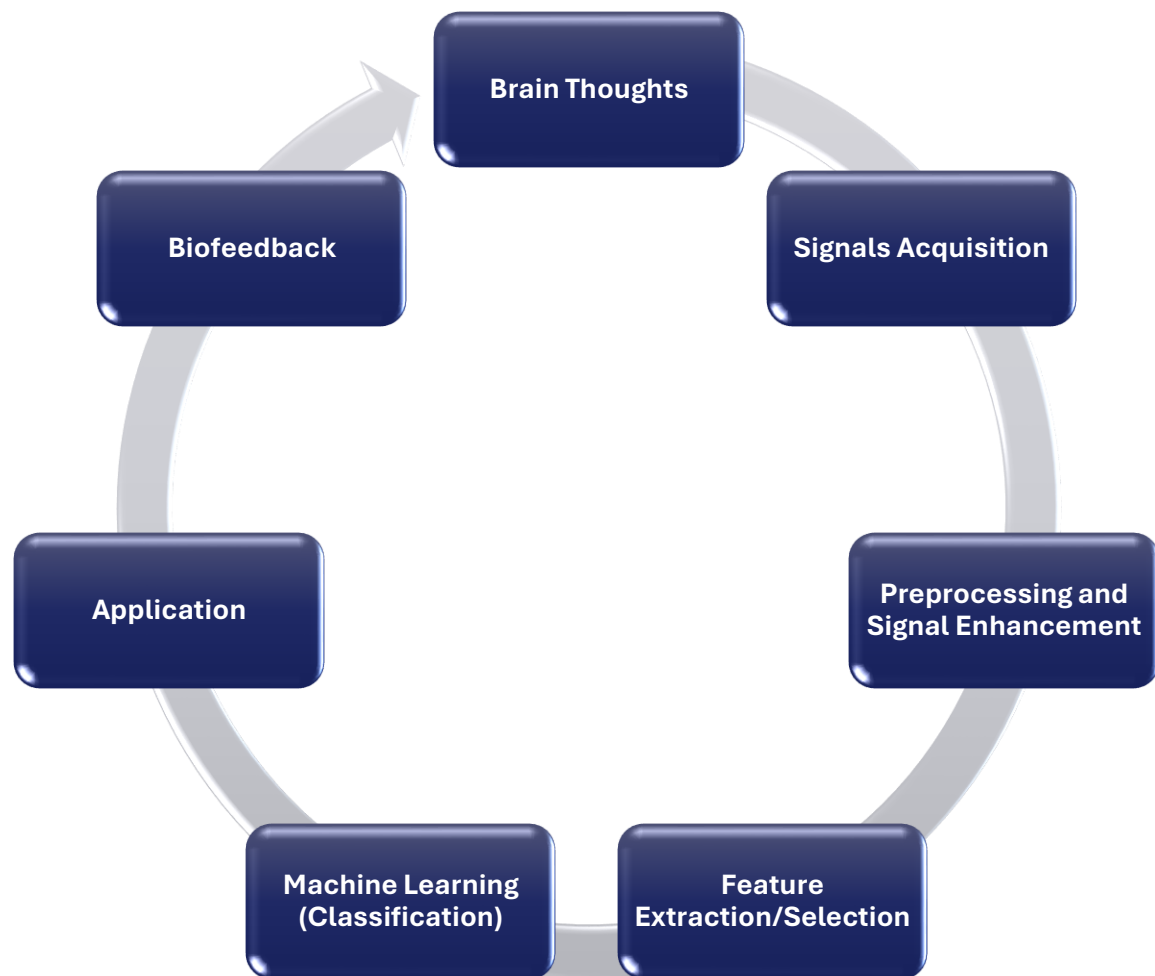
Restore

- Stroke rehabilitation
- Senses (e.g. vision)

Entertainment

- Home automation
- Computer games

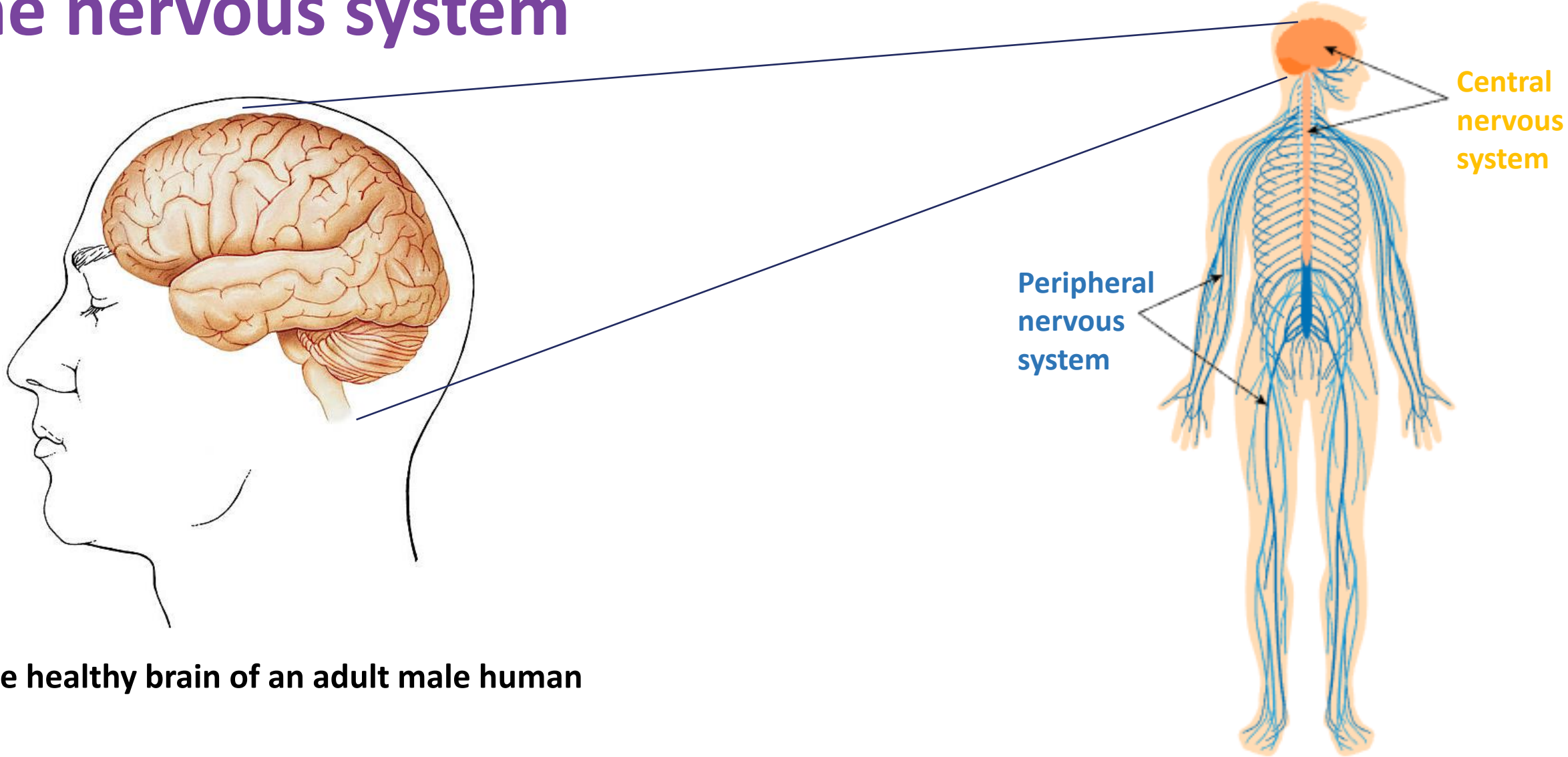
BCI system parts



BCI approaches

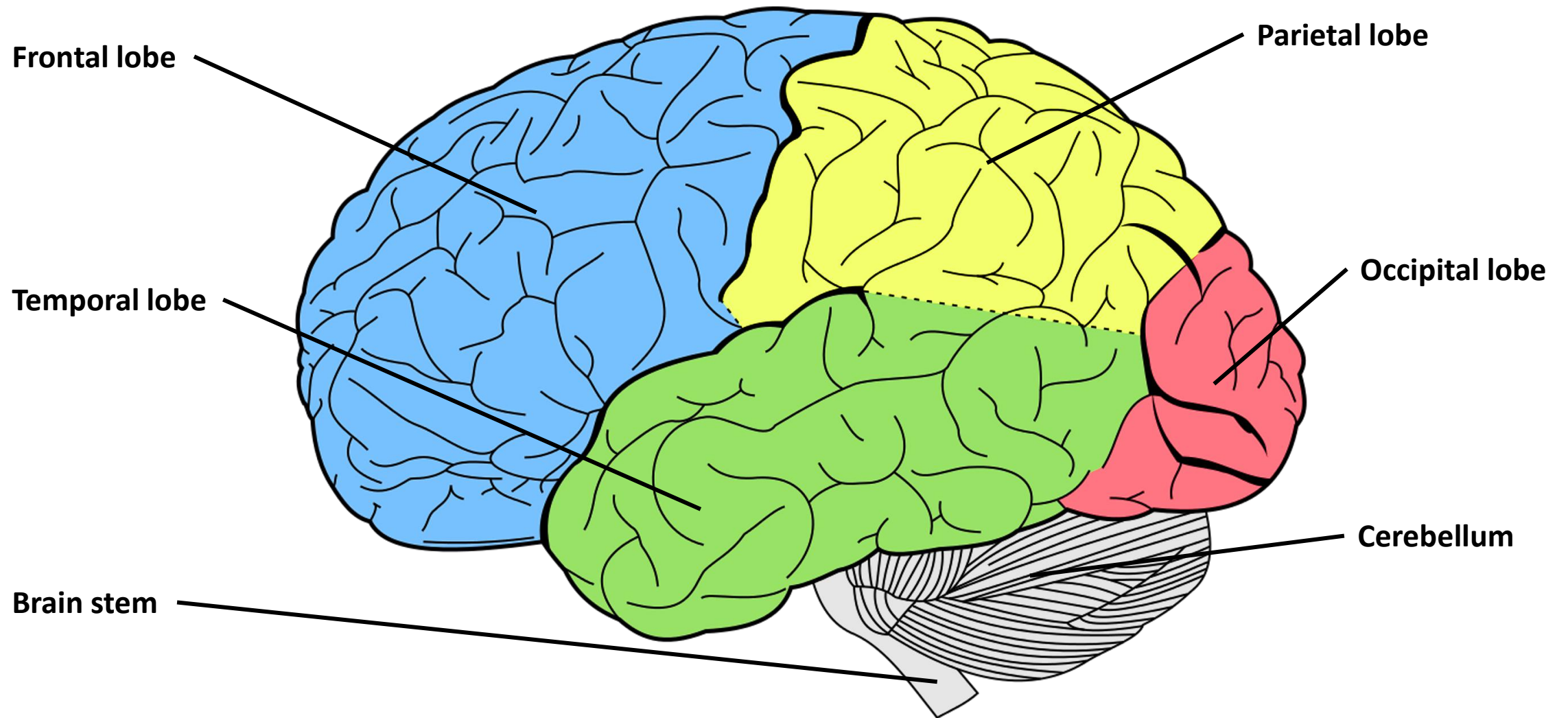
Categories		
User Active Control	Versus	User Natural Intent
Computer Operant Conditioning	Versus	Computer Pattern Recognition
Synchronous Detection (System-Paced)	Versus	Asynchronous Detection (User-Paced)
Offline Data Analysis	Versus	Online Data Analysis
Non-Invasive Neuroimaging	Versus	Invasive Neuroimaging
Universal/Multi Systems	Versus	Individual/Single Systems

The nervous system

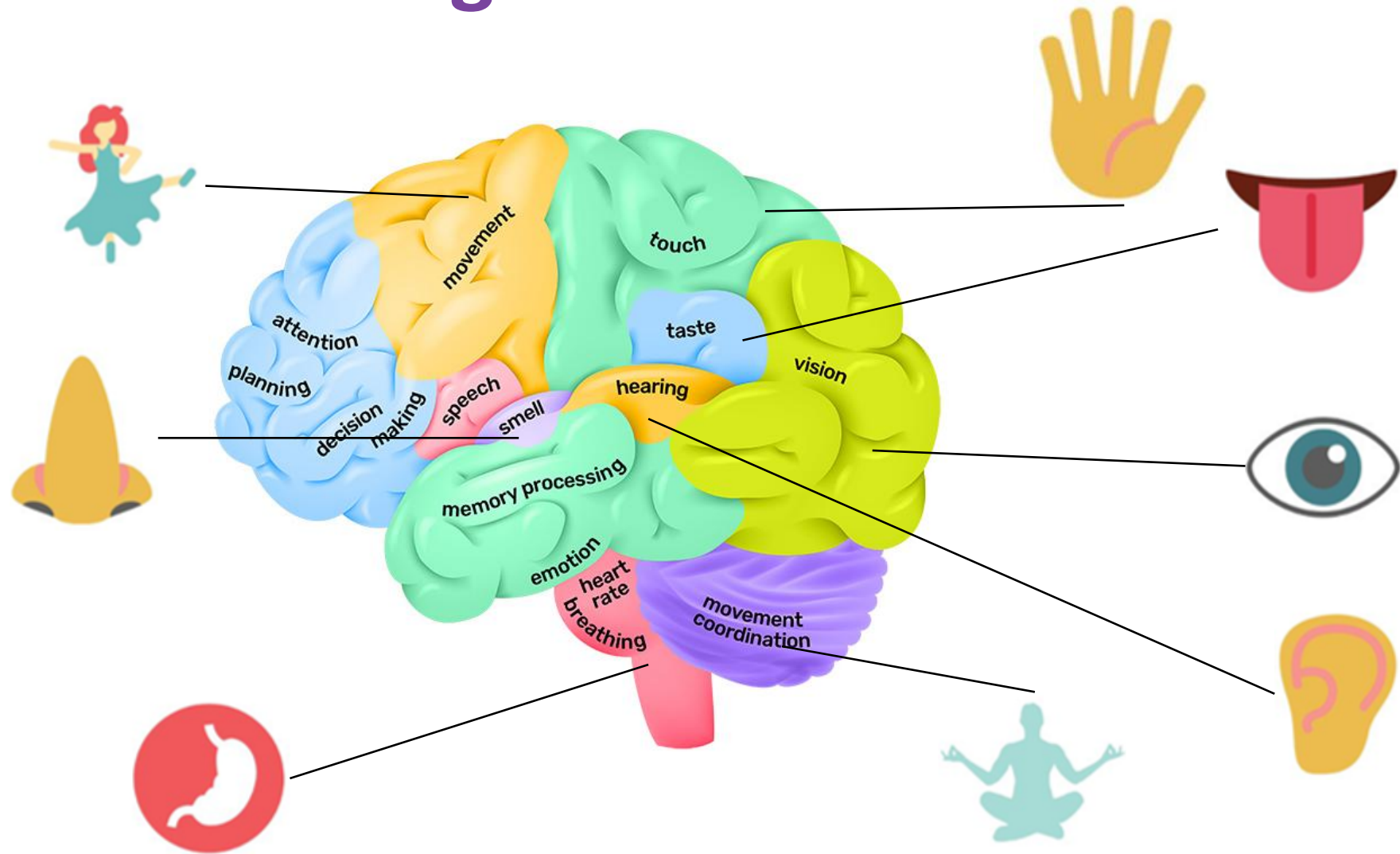


The healthy brain of an adult male human

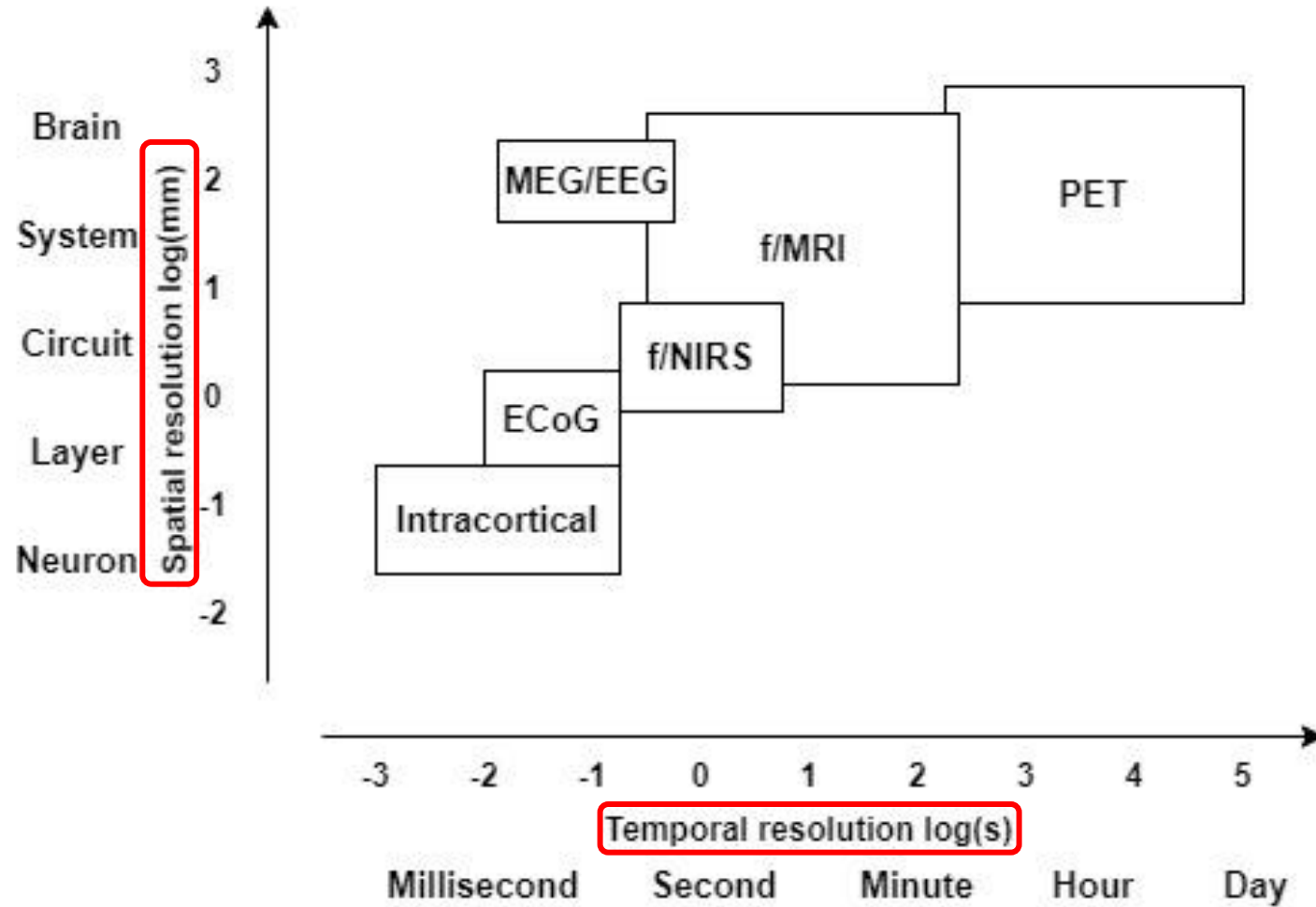
Brain anatomy



Basic functions vs regions



Neuroimaging



Electroencephalography (EEG)

EEG visualizes electrical brain activity noninvasively.

Neuronal activation generates a localized current flow and scalp electrical field.

The scalp electrodes' weak electrical signals must be amplified before being visualized.

EEG voltage must be compared to a reference point.





**MARKET RESEARCH
&
NEWS**

Your Growth. Our Insight.

Wireless EEG Headset Market

2024–2034

CAGR (2024–2034)

 **12.6%**

Market Size by 2034

 **USD 1.9 Billion**

Key Growth Drivers



Rising Neurological Disorders



Growing Wearable Neurotechnology



Expanding BCI Applications



Digital Health Investments

Top Companies

 **COMPUMEDICS**

natus
Natus Medical

 **CADWELL**
Cadwell Industries

 **BRAIN PRODUCTS**

ANT NEURO

neuroelectronics

EMOTIV

 **OpenBCI**

 **Wearable Sensing**

 **Advanced Brain Monitoring**

NeuroSky

 **Bitbrain**

Cognionics

 **g.tec**
medical engineering

 **EG** Electrical Geodesics

Micromed

 **MITSAR**

 **Neuracle**

 **Kernel**

 **Ripple Neuro**



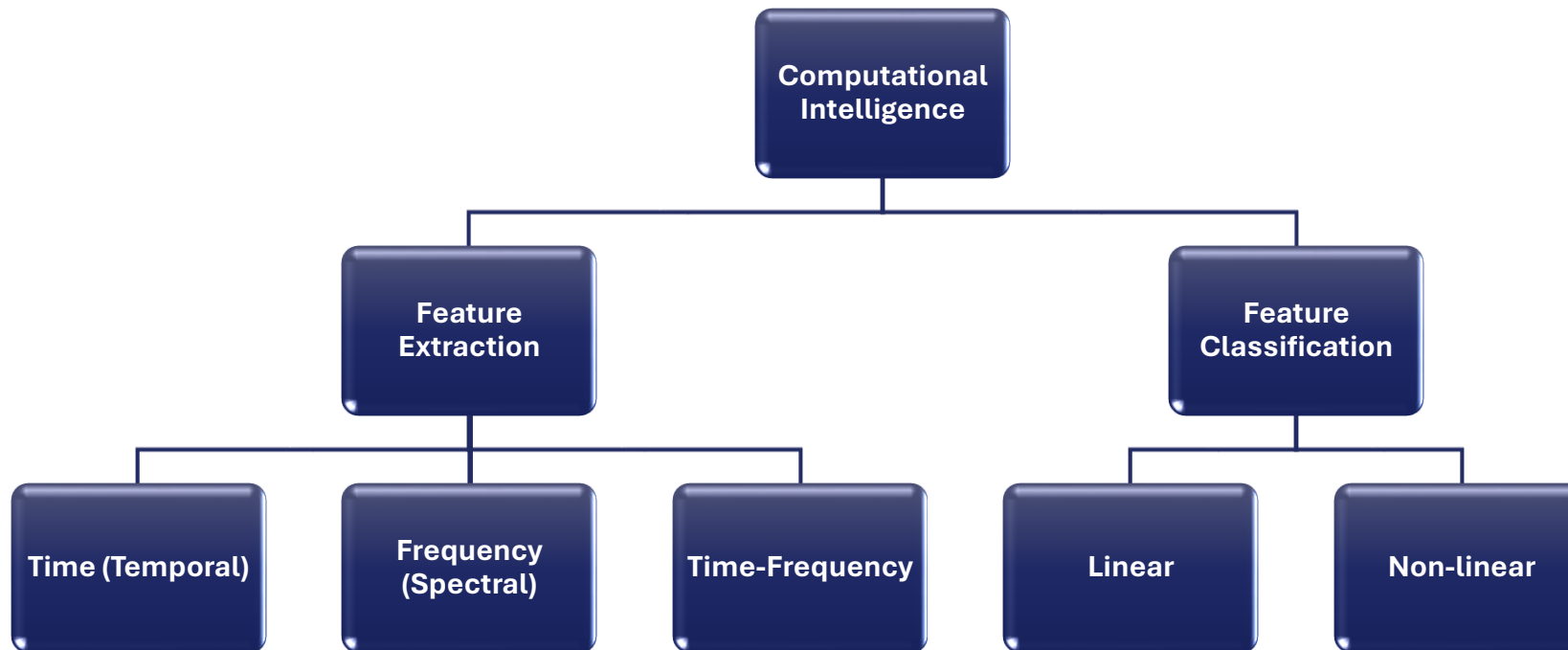
EEG Moves Toward Mainstream Hearables

Consumer EEG is moving from headbands and headsets toward integration in everyday earbuds, bringing sleep, cognitive-state, and control features closer to everyday use while raising new

**Even smaller
than an EEG
headset**

Computational intelligence for EEG-based BCI

The BCI uses computational intelligence to extract essential features from the pre-processed EEG signals and feed the feature vector into a classifier to control the BCI output.



Prompting the brain

NeuroBench studio brings together

- Conversational AI assistant
 - Ask questions about EEG data in plain language and interact with your analysis workflow.
- Visual MLOps pipeline builder
 - Build EEG workflows for dataset streaming, preprocessing, experimentation, and model training through a visual interface.
- Real-time streaming inference
 - Explore continuous EEG streams with interactive visualisation and foundation models such as LaBraM, SignalUEPA, EEGPT, BIOT, etc.
- EEGDash integration
 - Stream remote EEG datasets directly into your workflow.



Challenges of implementation

Hardware

**Experimental
design**

**Signal
denoising**

**Inter-subject
variability**

BCI illiteracy

BCI illiteracy refers to the inability of a person to effectively control a BCI system.

- This means they cannot generate the necessary brain signals to operate the device as intended.

Causes of BCI illiteracy

- **Individual brain differences:** People have unique brain patterns and abilities to modulate brain activity.
- **BCI system limitations:** Not all BCI systems are equally effective for everyone.
- **Insufficient training:** Proper training is crucial for BCI operation, and inadequate training can hinder performance.
- **Psychological factors:** Factors like motivation, anxiety, or stress can impact BCI control.

*Thank
You!*



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